

EX NO:	Design of Low Pass Filter using Windowing Techniques
DATE:	4.1.Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB.

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Input Parameters
```

```
fp = 1000;    % Passband cutoff frequency (Hz)
```

```
fs = 4000;    % Sampling frequency (Hz)
```

```
N = 50;      % Filter order
```

```
wc = fp/(fs/2); % Normalized cutoff frequency
```

```
%% FIR Low Pass using Rectangular Window
```

```
b_rect = fir1(N,wc,'low',rectwin(N+1));
```

```
disp('--- Rectangular Window Coefficients ---');
```

```
disp(b_rect');
```

```
%% Frequency Response
```

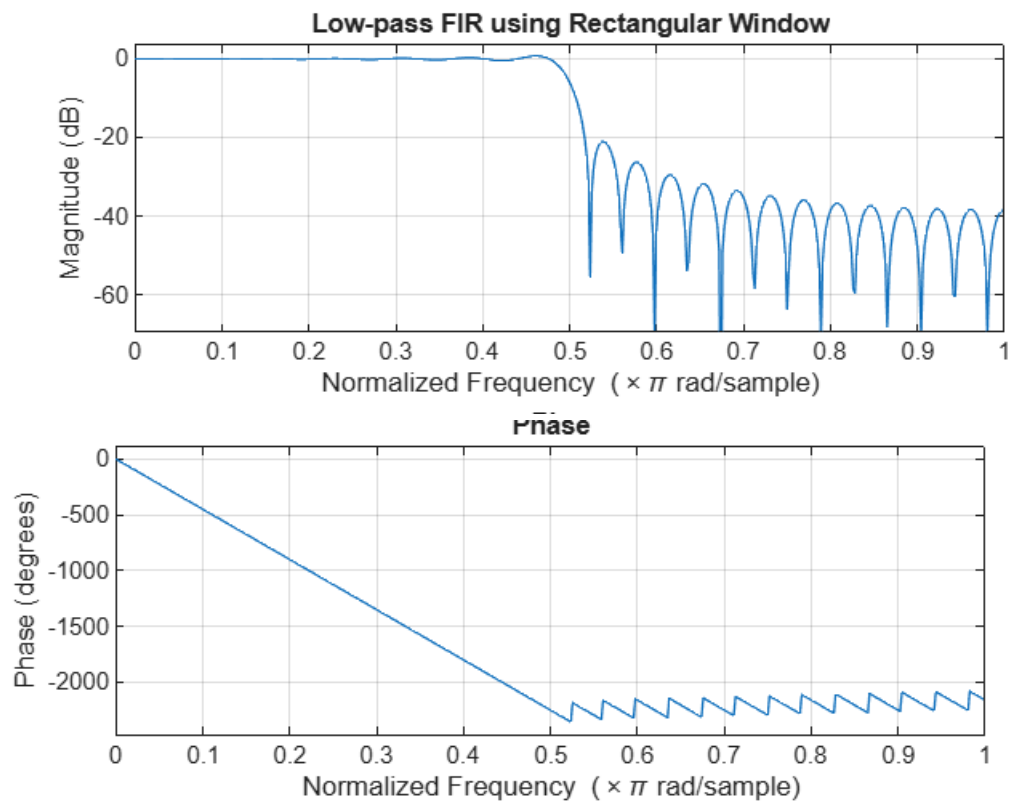
```
figure;
```

```
freqz(b_rect,1);
```

```
title('Low-pass FIR using Rectangular Window');
```

--- Rectangular Window Coefficients ---

```
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
```



Ex no	Design of Low Pass Filter using Windowing Techniques
date	4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

clc;

clear;

```

close all;

%% Input Parameters
fp = 1000;    % Passband cutoff frequency (Hz)
fs = 4000;    % Sampling frequency (Hz)
N = 50;       % Filter order

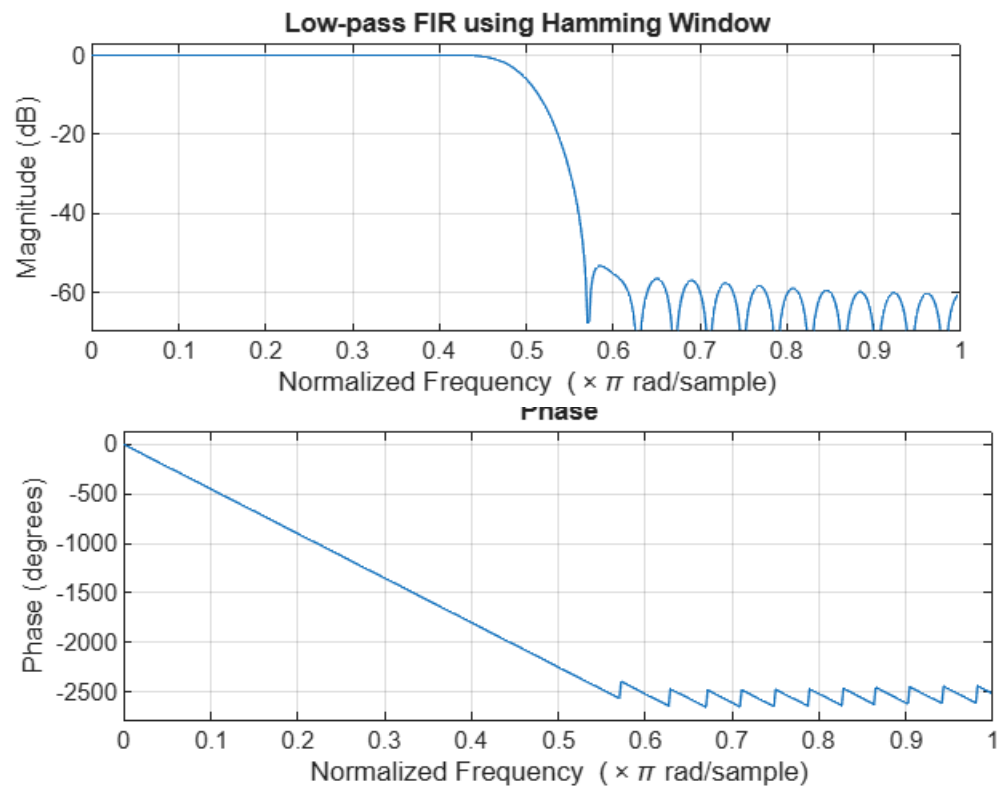
wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass using Hamming Window
b_hamm = fir1(N,wc,'low',hamming(N+1));

disp('--- Hamming Window Coefficients ---');
disp(b_hamm');

%% Frequency Response
figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');
    --- Hamming Window Coefficients ---
        0
    -0.0084
        0
     0.0125
        0
    -0.0181
        0
     0.0260
        0
    -0.0379
        0
     0.0580
        0
    -0.1026
        0
     0.3168
     0.4995
     0.3168

```



--- Hamming Window Coefficients ---

```

0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0

```

Ex no	Design of Low Pass Filter using Windowing Techniques
Date	4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

```

clc;
clear;
close all;

%% Input Parameters
fp = 1000;    % Passband cutoff frequency (Hz)
fs = 4000;    % Sampling frequency (Hz)
N = 50;       % Filter order

```

```

wc = fp/(fs/2); % Normalized cutoff frequency

```

```

%% FIR Low Pass using Hanning Window

```

```

b_hann = fir1(N,wc,'low',hanning(N+1));

```

```

disp('--- Hanning Window Coefficients ---');

```

```

disp(b_hann');

```

```

%% Frequency Response

```

```

figure;

```

```

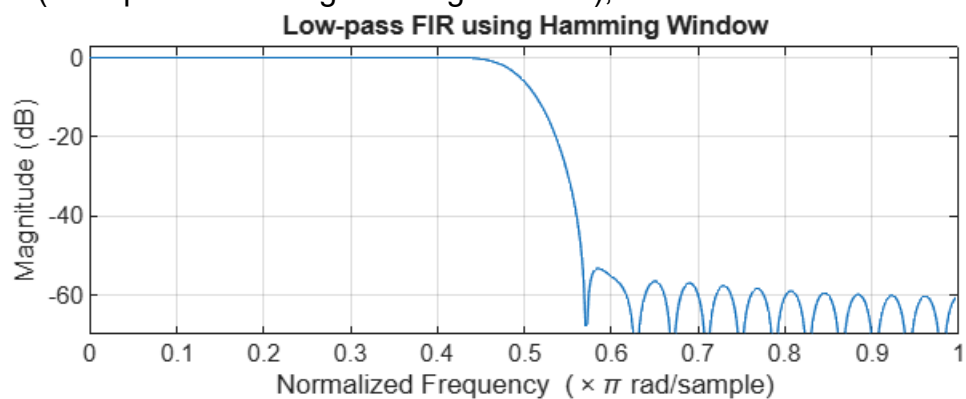
freqz(b_hann,1);

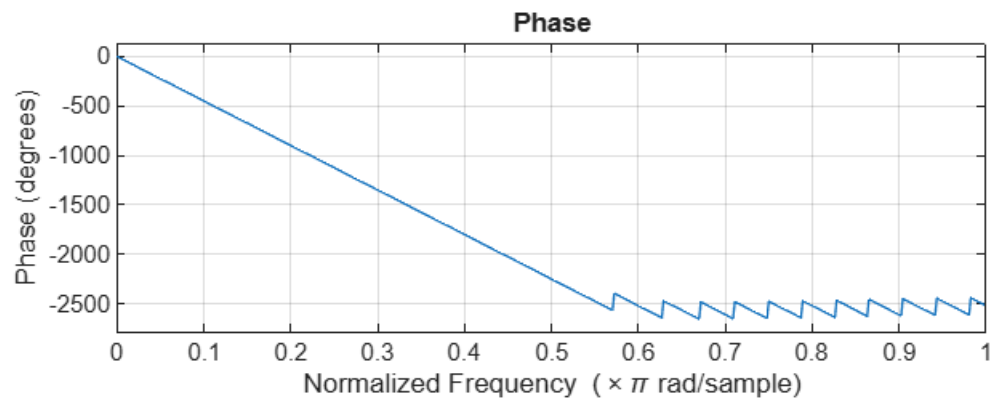
```

```

title('Low-pass FIR using Hanning Window');

```





--- Hamming Window Coefficients ---

0.0010	-0.0181
0	0
-0.0013	0.0260
0	0
0.0021	-0.0379
0	0
-0.0034	0.0580
0	0
0.0055	-0.1026
0	0
-0.0084	0.3168
0	0.4995
0.0125	0.3168
0	0
-0.0181	-0.1026
0	0
0.0260	0.0580
0	0

--- Hamming Window Coefficient

0

0.0260

0

-0.0181

0

0.0125

0

-0.0084

0

0.0055

0

-0.0034

0

0.0021

0

-0.0013

0

0.0010